

The World

MONDAY, JULY 6, 2026

INDIAN EXPRESS UPSC IAS EDITION HD 06~07~2026

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All the topics of this UPSC IAS Edition are directly or indirectly important for the prelims & main examination. There are some topics which can be coded in answer writing of other topics in the main exam.

Cargo ship under attack in Red Sea off Yemen, says British military



Associated Press
Cairo, July 5

A CARGO ship came under attack Sunday off the coast of Yemen in the Red Sea, the British military said, the latest maritime attack off the conflict-torn Arab country. The United Kingdom Maritime Trade Operations center reported the attack off the coastal city of Hodeida, which is under control of the Iranian-backed Houthi rebels.

The UKMTO said the ship reported being “under attack by unknown armed assailants” 30 nautical miles (55 kilometers) southwest of Hodeida.

A skiff approached the bulk carrier and opened fire, forcing the ship's security guards to return fire, before sailing back to a larger ship about 2 nautical miles (3.7 kilometers) away with its automatic identification system switched off, the UKMTO said.

The cargo ship and crew were reported safe, the British military agency said, adding that authorities were investigating. No group immediately claimed the attack. The Houthis have threatened to begin attacking ships again, though they have not been carrying out attacks. A Houthi spokesman did not respond to a request for comment.

The rebels previously fired drones and missiles at ships that pass by territory they control near the narrow Bab al-Mandab Strait at the southern end of the Red Sea.

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The Indian EXPRESS

JOURNALISM of COURAGE

NEW DELHI, LATE CITY, 20 PAGES

₹7.00 (₹8 RAIPUR, ₹15 SRINAGAR) • www.indianexpress.com

SAVES ON WATER, LABOUR COST

Direct-seeded rice gains traction among farmers amid El Nino shadow on monsoon



A farm worker transplants paddy seedlings at a field on the Delhi-Haryana border. HARISH DAMODARAN

Harish Damodaran
New Delhi, July 5

FOR BASMATI paddy grower Satyavan Sehrawat, the two biggest challenges are water and labour.

The traditional method of cultivating the cereal grain crop — by puddling (flooding the field and tilling repeatedly to break down soil clods, plug large pores and create a soft seedbed) and transplantation (of seedlings initially raised in nurseries) — is highly water-intensive.

“Puddling itself consumes water equivalent to three irrigations. For the first three weeks

after transplanting, the field has to be irrigated every second day to maintain a standing water depth of 4-5 cm. You need to irrigate even thereafter once a week,” said the 56-year-old Sehrawat, who farms basmati on 50 out of his 54-acre holding in Delhi’s Daryapur Kalan village.

Depending on the crop’s duration (115-120 days from seed to grain for Pusa Basmati-1509 and 140-145 days for Pusa Basmati-1121), and also how much it rains, the total irrigations can be anywhere between 20 and 30. Each irrigation uses over 200,000 litres of water per acre.

»CONTINUED ON PAGE 2

Elephant that faced off with tigers, patrolled Kaziranga laid to rest with guard of honour

Sukrita Baruah
Guwahati, July 5

ONE OF Kaziranga National Park's most iconic patrol elephants, 66-year-old Joymala, died on Saturday night and was laid to rest on Sunday with a guard of honour by Forest Department personnel in Kaziranga.

Stories about Joymala's 30 years of service as a patrol elephant are well known in the region.

In 2004, Joymala and its then 25-year-old mahout Satyabhanu Pegu were part of a team from the Assam Forest Department on the lookout for a large tigress that had killed cattle in a village close to the boundary of the national park. In a series of events well recorded and remembered among those who work in wildlife, the tigress at one point leapt towards Joymala's head while the elephant was carrying the range officer, a veterinarian, and a guard on its back, along with its mahout. A record of the incident by the Wildlife Trust of India says the tigress swiped at Pegu, injuring him, and did not land on Joymala, who stepped back. It says that as the tigress landed on the ground, Joymala tried to pin it

down with a foot, and the tigers struggled loose and ran away in the ensuing commotion.

A video of the incident, taken by the then divisional forest officer, RK Das, who was riding on another elephant just behind them, went on to be widely viewed.

Pegu, however, lost three of his fingers in the incident, bringing an end to his time as a mahout. However, Joymala continued as a patrol elephant for over 15 years, under the care of another mahout, Nilakhanta Koch, participating in anti-poaching patrols, wildlife monitoring, rescue operations, and routine forest protection duties.

“The elephants are not engaged in active duties after they cross 60 years of age, but she was in the camp post-retirement and was a motherly figure for younger calves. At least three of her own calves are currently patrol elephants in the national park,” said current DFO Arun Vignesh.

According to a statement by the national park, the elephant died on Saturday night at the Naloni area under Agorotoli Range of the national park, after being treated for illness for nearly a year.

El Nino

Those calculations matter, especially in an El Niño year, with rainfall across India being 24.1 per cent below-normal in the current monsoon season (June-September) until July 5.

But it isn't water alone. Transplanting paddy seedlings in puddled fields requires labour, which is getting increasingly scarce. "I pay Rs 4,000 per acre to the transplanting workers, plus arrange for their chai-paani (refreshments) and LPG cylinder or firewood. Smaller farmers who cannot provide these have to shell out Rs 5,000 per acre," noted Sehrawat.

That's where DSR or direct-seeded rice technology comes in. It dispenses with preparation of nurseries (where paddy seeds are sown and grown into young plantlets over 25-30 days) and uprooting of the seedlings for transplanting in the main field.

(ten times the nursery area). Nor is there need for puddling and flooding of fields.

In DSR, the paddy seeds are directly sown in the main field, just like wheat. In transplanted paddy fields, the standing water acts as a natural herbicide, depriving weeds of oxygen and sunlight for their seeds to germinate or already-emerged seedlings to survive. Continuous flooding is necessary to prevent weed growth, especially during the crop's early stage.

DSR basically replaces water with chemical herbicides. It means growing paddy varieties or hybrids that can “tolerate” their application — in this case, of Imazethapyr, a herbicide that controls a wide spectrum of grassy and broadleaf weeds.

The normal Pusa-1509 and Pusa-1121 varieties Sehrawat cultivates cannot “tolerate” Imazethapyr, as the chemical does not distinguish the crop from the weeds and ends up killing both. What he is, however, growing now are Pusa Basmati-1985 and

Pusa Basmati-1979. These are the Imazethapyr herbicide-tolerant (HT) versions of the same popular basmati varieties bred by the Indian Agricultural Research Institute (IARI) in New Delhi.

The HT varieties have been developed through non-genetically modified mutation breeding, involving altering the ALS (acetolactate synthase) gene. This is a gene naturally present in rice plants and not introduced from soil bacteria and other unrelated species (as with genetically-modified Bt cotton or hybrid mustard).

“The ALS gene codes for an enzyme responsible for the synthesis of leucine, isoleucine and valine, which are essential amino acids for plant growth and development. Imazethapyr sprayed on normal paddy binds itself to the ALS enzyme, inhibiting the production of these amino acids,” explained A K Singh, former IARI director.

The two HT basmati varieties bred by him and fellow

IARI scientists contain a mutated ALS gene whose DNA sequence has been altered. While amino acid synthesis isn't inhibited, the ALS enzyme no longer has binding sites for Imazethapyr. The paddy plants can then "tolerate" the herbicide, which will only kill the weeds.

Last year, Sehrawat tried out Pusa Basmati-1985 and Pusa Basmati-1979 on one-acre land each. The results were good enough for him to expand DSR to 20 acres this year: 16 acres under the first and four under second HT variety.

"There's no paneeri (nursery preparation) here or labour required for ropai (transplanting). And I am saving 30-35 per cent water, as the field does not have to be flooded. Irrigation is given just to retain moistness," he pointed out.

In 2025, the total area under the two HT basmati varieties was estimated at 10,000 acres. A K Singh sees that touching 100,000 acres this year.

Odisha SIR: 20 lakh deletions in draft rolls; Opp claims genuine voters were removed

Sujit Bisoyi
Bhubaneswar, July 5

NEARLY 20.14 lakh names of electors have been identified for deletion in Odisha, according to the draft electoral rolls published on Sunday, following a month-long enumeration process as part of the Special Intensive Revision (SIR) of the state voters' list. While the number of electors before the SIR was 3.33 crore, it has come down to 3.13 crore in the draft rolls. This includes 1.60 crore male and 1.53 crore female electors.

The Opposition BJD raised concerns over the draft electoral roll, saying more than 10,000 voters were identified for deletion from 75 of the total 147 Assembly constituencies in the state. "Many genuine names were excluded in the draft electoral roll because of minor anomalies, which need to be included during the claims and objection process," said senior BJD leader Debi Prasad Mishra.

Among the names deleted from the draft rolls, 8.32 lakh were found to be deceased. While 10.07 lakh names were deleted because they had moved elsewhere or were absent during the process, 1.58 lakh were found to have been enrolled in multiple places.

Odisha's Chief Electoral Officer R S Gopalan, said the claims and objections period starts from Sunday. The final roll, will be published on September 6. "If anyone's name is not found in the draft voter list, he/she has to fill and submit a Form 6 with a declaration form and other documents to get the name included in the final roll," said Gopalan.

"No name can be deleted without notice and a speaking order by ERO/assistant ERO. Any aggrieved elector may appeal to District Collector and thereafter to CEO under the relevant sections of the Representation of the People Act, 1951," he said.

19.34 lakh in Manipur draft rolls, 1.58 lakh did not submit forms

Express News Service
Imphal, July 5

THE MANIPUR draft voters' list, published on Sunday, contained 19.34 lakh electors of the state's 20.93 lakh or 92.42% across 60 Assembly constituencies, under the Special Intensive Revision (SIR) exercise conducted between May 30 and June 28, said Chief Electoral Officer (CEO) Arun Kumar Sinha.

A total of 1.58 lakh electors (7.58%) did not submit their enumeration forms during the SIR process, the CEO said. Of the 19.34 lakh electors, 9.40 lakh are male, 9.93 lakh female and 2.93 third-gender, according to the draft rolls. 20,442 electors

are in the 18-19 age group.

Sinha assured that no elector's name would be deleted from the draft rolls without due process. "Nobody can remove anybody from the draft electoral roll unless he has been given an opportunity of hearing. Only after hearing and passing a speaking order, can a person's name be deleted from the draft electoral roll."

Even after the publication of the final voters' list on September 6, aggrieved electors can appeal under the provisions of the Representation of the People Act, 1950. Those whose names are missing from the rolls for failing to submit enumeration forms can still apply for inclu-

sion by filing Form-6, along with the prescribed declaration, during the Claims and Objections period from July 5 to August 4, 2026, Sinha said.

He said that the Notice and Verification Phase will continue from July 5 to September 2, during which EROs will scrutinise claims and objections and issue notices, wherever required. He said electors whose records require verification will be given an opportunity to produce the necessary documents before a decision is taken. For correction of particulars such as name, address and change of Assembly constituency, electors may submit Form-8.

Govt issues notice to Meta on Insta ads linked to child sexual abuse

Soumyarendra Barik
New Delhi, July 5

THE MINISTRY of Electronics and Information Technology (MeitY) has issued a notice to Meta-owned Instagram, directing the social media platform to disable all advertisements and content promoting and facilitating access to child sexual abuse material (CSAM).

The government has also asked Meta to provide a detailed explanation, within a week, on why and how the platform had allowed users to post paid advertisements containing such material.

The notice was sent Saturday evening after BBC reported that Instagram was allowing paid advertisements promoting child sexual abuse material in India.

“The government has issued a stern notice to Meta on the issue. MeitY has ordered Instagram to disable all ads and

content promoting and facilitating access to CSAM, and sought a detailed explanation within seven days,” a senior government official said.

A Meta spokesperson said the company has a “zero-tolerance” policy on sharing such content, and that Instagram deploys artificial intelligence (AI) tools to proactively detect it.

“Meta has a zero tolerance policy for soliciting or sharing CSAM, including in ads. We use advanced AI technology to proactively detect violating content and individuals, but we are in a constant battle with criminals who hide among our 3.5 billion users and try to evade our detection. That is why our expert teams are constantly working to improve our defences, develop new technology to root out predators, block links to violating websites, and share intelligence with other companies so they can take action too,” the spokesperson said.

The episode has ignited an important conversation surrounding safe harbour available to social media companies, under which they are allowed legal immunity from hosting third-party, user-generated content on their platforms. The key idea behind the immunity is that companies do not have prior knowledge of what users might post on their platforms, and as a result cannot be held liable for them until such content is brought to their notice, under a legal provision called “actual knowledge”.

However, the government is now considering whether such protections should be available to companies for hosting advertisements, given that there is a preliminary moderation that platforms are expected to carry out while reviewing advertising material.

“It cannot be a case where a social media platform is allowed to host such deeply

problematic advertisements, and have no consequences due to safe harbour protections. We are discussing whether advertisements on social media should have the same protections as regular content,” an official said.

Broadly, many are also seeing the development as a potential precursor that could kickstart conversations within the Centre around regulatory protections for children on social media.

Countries like Australia, the United Kingdom and the United Arab Emirates, among others, have moved to ban social media access for children. India is learnt to have discussed a graded approach to such limits but a concrete step is yet to be taken, The Indian Express had reported earlier.

This is the second instance within a week when Meta has found itself in the crosshairs of the Indian government.

Earlier, the government asked Meta-owned WhatsApp to not roll out its upcoming username feature over concerns of impersonation and fraud. In a notice issued to the company, the IT Ministry gave WhatsApp three days to furnish a detailed explanation on the username feature.

In its notice, the government is understood to have told WhatsApp that the feature may materially increase the incidence of online fraud, phishing, digital arrest scams and impersonation attacks, by enabling bad actors to solicit and message unsuspecting users. The government has also flagged that the feature may facilitate impersonation of individuals, public authorities, financial institutions and government agencies, by permitting the adoption of usernames closely resembling those of genuine people and institutions.

• BRIEFLY



Indian Navy's Sail Training Ship *Sudarshini* at the US International Naval Review and Sail celebrations in New York. PTI

250th ANNIVERSARY OF U.S.

INS *Sudarshini* participates in maritime spectacle on Hudson River

INDIAN NAVY'S sail training ship — INS *Sudarshini* — carrying the tricolour sailed into the New York port and past the Statue of Liberty on July 4, joining vessels from the US and several other countries for a grand event marking the 250th anniversary of the US' independence. The ship's presence in the International Naval Review and Sail 4th 250 celebrations hosted by America, also reflects the “enduring friendship, mutual respect, and expanding maritime partnership between India and the United States”, the Indian Navy said on Sunday. The visit is part of the ship's 10-month transoceanic expedition, Lokayan 2026. PTI



El Niño, Iran sharpen food prices challenge

THE SOUTHWEST monsoon was 39.8 per cent below normal in June, with the all-India average rainfall of 99.5 mm the fifth lowest recorded for that month after 2014, 2009, 1926 and 1905. This month, which coincides with the peak plantings of the kharif crop, has been better so far. Not only has the cumulative rainfall deficit for the season (June-September) narrowed to 24.1 per cent as on July 5, the monsoon has advanced to cover roughly 95 per cent of the country's area. That should, in turn, help boost kharif sowing acreages, which were overall 22.7 per cent lower compared to last year's coverage till June 25. The lag was even more for oilseeds (53.3 per cent), pulses (30.5 per cent) and cotton (34.6 per cent). Those gaps can potentially close with the monsoon's revival from June 30.

But the outlook, on the whole, is still not great. The India Meteorological Department has forecast average rainfall over the country for July, too, to "most likely to be below normal". Worse, the dreaded El Niño is in a "moderate" phase now, while predicted to intensify into a "strong" event during the second half of the monsoon season and turn "very strong" over October-January. The impact wouldn't be limited to the monsoon rainfall and the kharif crops grown during this season. Since El Niño is known to suppress rainfall and also raise temperatures, it could result in a relatively short and warm winter, affecting the upcoming rabi season crops like wheat, mustard, *chana*, *masoor* and potato as well. The last "strong" El Niño event of 2023-24 led to annual retail food inflation averaging over 8.5 per cent between July 2023 and December 2024. Without proactive supply-side management, there is every risk of a repeat of that protracted episode of elevated food prices.

India imported a record 16.9 million tonnes (mt) of vegetable oils in 2025-26 and 7.3 mt of pulses the previous fiscal. El Niño — which has overtaken Iran as the No 1 risk factor for the Indian economy today — will make further imports inevitable. While keeping the import window open, the government must ensure minimum support prices for pulses, oilseeds, millets and cotton through payment of the difference over open market rates. It will incentivise farmers to cultivate these crops, instead of water-intensive rice, wheat and sugarcane. The year 2026-27 will also test the flagship crop insurance (PMFBY) and rural employment (VB-G RAM G) schemes. Their implementation on the ground matters as much as the supply-side measures to control food inflation.



OVER THE BARREL
BY VIKRAM S MEHTA

THE FOG of conflict in the Middle East has lifted somewhat, and one can now discern the geoeconomic and geopolitical changes it has wrought. Whether these changes dissolve over time and conditions revert to the status quo ante February 28, the day American bombers took out the leadership of Iran, or calcify into an invariant structural shift remains to be seen. Irrespective, petroleum-import-dependent countries like India should factor these changes into their discussions on energy and national security.

In this article, I identify four such changes.

First, diminishing confidence in America as a guarantor of security. Other than the most hardened MAGA acolyte, everyone agrees that America was worsted by Iran in this conflict. It failed to achieve any of its stated objectives (regime change, elimination/dilution of Iran's cache of weapons-grade uranium and degradation of Iran as a regional power) despite, according to its Central Command, American bombers flying 10,000 sorties, hitting 1,30,000 targets inside Iran, destroying 85 per cent of Iran's missile manufacturing capabilities and 70 per cent of its missile launch infrastructure and sinking most of its navy between February 28 and April 8, the date of the first ceasefire.

Moreover, America failed to protect its Gulf allies from Iranian missiles and drones. Saudi Arabia, the UAE, Qatar, Bahrain and Kuwait all suffered physical and economic damage during the conflict. These countries had permitted America to establish military bases in their territory in exchange for the assurance of such protection. US Secretary of State Marco Rubio recently toured the region to smooth ruffled feathers, but more than anything he may have said, I suspect it was the words of his predecessor, Henry Kissinger, that resonated with the Middle East leaders: "It may be dangerous to be America's enemy, but to be America's friend is fatal". This dent in confidence opens up space for non-Western countries to deepen their security relationships in the region and, in particular, creates an opportunity for India's defence-related manufacturing and technology companies to explore mutually beneficial opportunities.

Second, the emergence of "choke points" as a lever of power. The Strait of Hormuz, through which 20 per cent of global oil supplies flow, has long been recognised as a potential source of market disruption, but essentially only in analysis. Conventional wisdom has held that Iran does not have the capability to enforce a shipping blockade. Indeed, the one time it sought to do so in 1988, the US 5th Fleet blew up two Iranian frigates, demolished oil platforms and cleared the sub-sea mines within days. Oil tankers were barely impeded. This wisdom has now been upended. Through a combination of missile threats, attacks by small but fast speed boats and adroit social messaging, Iran did halt shipping traffic. The economic and political impact was a

India can reduce its exposure by expanding the storage capacity for strategic petroleum reserves from the current roughly eight days to at least 30 days, or better still, 60 days.

sharp rise in oil prices — it peaked at \$126 per barrel in April, up from \$75 per barrel before the commencement of bombing — imposition of demand-reduction measures across Asia, the drawdown of strategic reserves and a political backlash. One reason President Donald Trump sought peace was concern that the rise in gasoline prices at the US retail forecourt would hurt his party's electoral prospects in the November mid-term elections and render him a lame duck president for the balance of his tenure.

Iran's success in leveraging its geographic position for military advantage has focussed attention on "choke points" as a source of asymmetric technological, competitive and strategic power. The former UK PM, Rishi Sunak, has in an article advised countries/corporates not to seek control over the entirety of an integrated technology value chain but to create a niche position that can give them an "unassailable edge". He cites Arm Holdings, whose chips are in 99 per cent of all smartphones, TSMC, the Taiwanese company that makes 70 per cent of all advanced semiconductor chips, and ASML, the Dutch manufacturer of extreme ultraviolet lithography machines, as successful examples of this approach. He could have added China's dominance of rare earths to this list.

Clearly, not every country/corporate can create such a choke point. Those who cannot must find ways of reducing their vulnerability. The Gulf countries are expanding/building overland pipelines as an alternative export route to the choke point of Hormuz. India can reduce its exposure by expanding the storage capacity for strategic petroleum reserves from the current roughly eight days to at least 30 days, or

better still, 60 days.

Third, OAPEC is now a spent cartel. The UAE has left it; several other countries are reportedly on the verge of leaving, and Iran is at military loggerheads with the Arab members. Had anyone asked experts to forecast the price of oil in the event of closure of the Strait of Hormuz, they would have indicated a range between \$150 and \$200 per barrel. In fact, other than the one-day peak of \$126/bbl, prices ranged in the \$90s. The reason for this relatively moderate trajectory was the reduction of imports by China from 11.4 mbd in February to 6.4 mbd in May, the drawdown of strategic reserves by the IEA and increased production from non-OPEC producers. Together, these factors illustrated the cartel's weakening influence.

Finally, the reemergence of Iran as a "responsible" state actor. I am no expert on Iranian politics, but from what I have read, the balance of power in the country is shifting towards leaders who, whilst Islamic, are not radical, nor ideologically driven to defend the principles of the 1979 revolution (notwithstanding the week-long funeral planned for Ayatollah Khamenei). They are technocratic, pragmatic and nationalist. They recognise public discontent cannot be indefinitely contained, and policy focus must now be on economic growth.

India has civilisational connections with Iran. Before the US sanctions, New Delhi received oil supplies from Tehran on superior credit terms. It should support Iran's developmental efforts with technical, manpower and financial assistance. This will not only strengthen ties but also enhance the probability of preferential access to energy.

The writer is chairman and distinguished fellow, Centre for Social and Economic Progress

Rice-to-ethanol policy is deepening economic and environmental stress



FROM PLATE TO PLOUGH

BY ASHOK GULATI, RITIKA JUNEJA AND PURVI THANGARAJ

INDIA PRODUCED 154 million metric tonnes (MMT) of rice in 2025-26, surpassing China to become the world's largest producer. India is also the top rice exporter, with exports of 24.5 MMT in 2025. This accounts for roughly 40 per cent of the global rice trade (61.3 MMT), exceeding the total exported by the next four largest exporters — Vietnam, Thailand, Pakistan and Cambodia. India also gives free food (Rs 5kg/person/month) to about 800 million people under the National Food Security Act (NFSA), of which almost two-thirds is rice. Even after this, the Food Corporation of India (FCI) held rice stocks of almost five times the buffer norm as of April 1. In 2024-25 (FY25), the carrying cost of this extra buffer stock was Rs 10,712 crore. The government has decided to divert significant quantities of broken/damaged rice to ethanol production to reduce these stocks and cut down carrying costs. In FY26, almost 5 MMT of rice has been used to produce ethanol. The economic cost of rice to FCI in FY27 is likely to be Rs 44/kg, while it is being given to ethanol plants at roughly Rs 23/kg, renewing the food-vs-fuel debate and raising questions about who is subsidising whom.

How did India, a country that lived from “ship to mouth” in the mid-1960s, become the world's largest rice producer, exporter, free distributor, and ethanol maker from the grain? The Green Revolution's success is only one part of the story. The other, more recent and disturbing, is the story of subsidies, which are wreaking havoc not only on the fisc but also on the environment, through higher greenhouse gas (GHG) emissions, soil degradation, groundwater contamination and biodiversity loss. Let us delve deeper and explore how India can rationalise rice pol-



ILLUSTRATION: C R SASIKUMAR

icies for a financially and environmentally sustainable future.

Three interlocking policies seem to be responsible for this situation. First, the open-ended procurement at the minimum support price (MSP). Although the MSP for paddy is Rs 2,369/quintal (2025-26), states often outbid each other. Chhattisgarh pays about 40 per cent above MSP, Telangana about 20 per cent, while others, such as Andhra Pradesh, Odisha, Punjab and Haryana, have their own variants of this competitive largesse either in the form of input subsidy or bonus on top of MSP. Second, free or near-free power across the rice belt. Since rice is a flood-irrigated crop that demands up to 25 irrigations per season, free power is effectively a licence to mine groundwater without consequence. Third, urea is sold at a price that bears no relation to its cost. The retail price of urea has been frozen at Rs 242 per 45-kg bag for years, with the government absorbing 85-90 per cent of the economic cost. India's own landed import price for urea had spiked to \$935-\$959/tonne in May

The ethanol blending programme is not a bad idea, in itself — it involves at least 20 per cent blending. However, mandating that ethanol producers source a fixed share of feedstock from FCI rice is counterproductive

— thanks to a modicum of cooling in the Strait of Hormuz conflict, the price now has dropped to a little less than half this amount. Yet, this huge arbitrage drives chronic over-application of urea on paddy fields as well as diversion to other industries and across borders. According to an ICRIER study, the combined subsidies (power, fertiliser and canal water) for paddy cultivation in Punjab amounted to Rs 38,973/hectare in 2023-24. These distorted and politically motivated policies of bonuses over MSP, coupled with highly subsidised power and fertilisers, have resulted in FCI's stocks overflowing. Rather than fixing the distorted incentive structure that has created such piles of rice, the government has chosen a second exit valve: Pour it into ethanol distilleries.

The ethanol blending programme is not a bad idea, in itself — it involves at least 20 per cent blending. However, mandating that ethanol producers source a fixed share of feedstock from FCI rice is counterproductive. This is like going back to Soviet-era con-

trols. Bureaucracy and politicians love controls as they benefit from rent-seeking. Industry wants freedom to choose the most efficient feedstock — whether maize, sugarcane, or rice — which government policies don't permit. So, the two strike a deal — rice, whose economic cost to FCI is Rs 44/kg, is sold to ethanol plants at Rs 23/kg on the justification that it is broken and damaged.

Production of a kilogram of rice requires, on average, about 4,000 litres of irrigation water. Even if one assumes half of it percolates back to aquifers, the other half is absorbed partially by the plant, and the remaining evaporates. Transplantation, the dominant practice in rice cultivation, emits large amounts of methane, which is 25 times more potent than carbon dioxide. And when states like Punjab and Haryana use almost 250/kg of fertiliser per hectare, they emit almost 5 tonnes of carbon dioxide/ha as the plant absorbs not more than 35-40 per cent of nitrogen — the rest goes into the environment as nitrous oxide, which is 273 times more potent as a GHG than carbon dioxide, or leaches into groundwater, contaminating it with nitrate. The contamination has been linked to blue baby syndrome, thyroid, diabetes and increased cancer risk. Why are we imposing such hazards on ourselves?

This must change. Otherwise, a boon of technology will become a bane of humanity. Wisdom lies in changing the course of the MSP policy. No bonuses, and limit the procurement in all states to not more than 40 per cent of its production. No compulsion on ethanol plants to use rice from FCI. Maize is a much better option for ethanol production. Limit the free supplies under PDS to only the most vulnerable — antyodya — and charge the rest at least half the MSP. Promote direct-seeded rice to conserve water. And above all, fix fertiliser-related issues by moving from subsidies to income support to farmers and decontrolling fertiliser prices. If the Narendra Modi government can implement such reforms, it will do a great service to India's farmers and our planet.

Gulati is distinguished professor, Juneja a senior fellow, and Thangaraj a consultant at ICRIER. Views are personal



KIRIT S PARIKH

BY THE end of June, oil prices had come down to \$74 per barrel, the pre-crisis level of early February. I must confess to being pleasantly surprised by the fall in prices. Iran's bombing of the Middle East's oil infrastructure made me think it would take many months for global supply to recover and crude prices to fall to pre-crisis levels. However, price is an outcome of both supply and demand. With high prices, some reduction in consumption can be expected. But the demand for Gulf oil has also fallen because countries diversified their sources, and took measures to enhance long-term oil security.

India has managed the crisis very well. Supply was ensured. Hardly any retail outlet ran dry. Domestic consumption was reduced with 20 per cent ethanol blending. Electric mobility was pushed. Consumers were protected: Prices were not increased in step with import prices, and oil marketing companies and the government bore a large part of the burden. This was justifiable if the government expected the crisis to last no more than three to four months. Fortunately, it was proved right. The revenue forgone was Rs 1.7 lakh crore. The consumer price of petrol was later increased by Rs 3 per litre, around a 7 per cent increase. The price increase is by far the smallest among major oil-importing countries.

More than half of the LPG consumed by Indian kitchens would arrive through the Gulf, and that suddenly stopped. Government ac-

Supply was ensured. Domestic consumption was reduced with 20 per cent ethanol blending. Consumers were protected: Prices for them were not increased in step with import prices

tion was decisive. An LPG control order, issued within eight days of the disruption, directed domestic refineries to increase production. Within seven days, production increased from 35,000 to 54,000 tonnes per day. The regulated price of LPG was kept at Rs 942, and for the beneficiaries of the Pradhan Mantri Ujjwala Yojana at Rs 642, with a direct benefit transfer of Rs 300 per month to nearly 10.6 crore households. Commercial LPG supply was restrained for some time, and industrial and commercial users were encouraged to use other fuels.

The management of supply has been built on infrastructure developed over the last decade or so. Between 2011 and 2022, LPG import terminals doubled, to 22, increasing the country's LPG import capacity from 12 MMTPA to 32.3 MMTPA. At the same time, the city gas distribution network coverage increased from 55 cities to 300-plus cities. From 2011 to 2022, LNG import terminals doubled, and India expanded its crude oil sourcing network from 27 to 41 countries, which facilitated supply diversification and inter-fuel substitution.

India was also able to get a number of its ships through the Strait of Hormuz despite the blockade. This required coordination between more than one institution and ministry. The Ministry of Petroleum and Natural Gas identified and shared the list of priority vessels carrying India-bound cargo with the Ministry of Ports, Shipping

and Waterways. The latter compiled additional vessel-specific information and forwarded it to the MEA, which liaised directly with the Iranian embassy in New Delhi to seek clearances, while simultaneously instructing the Indian embassy in Tehran to expedite the processing of such requests by engaging with Iranian authorities. The Navy then coordinated with the vessels to ensure their safe passage through the Strait of Hormuz. This approach ensured that requests were pursued through both capitals at once, reducing the turnaround time for approvals.

India has long been concerned about disruptions in oil and gas supply from the Gulf. Its bilateral engagements with several countries, which predate the crisis, facilitated the diversification of imports. India continues exploration efforts domestically as well as beyond the country's shores and has procured oil and gas assets abroad. Thus, in a sense, the crisis has made India a little more energy secure.

However, the challenge for the government is not over. Consumers have been protected at a high cost to the government and oil marketing companies. The government faces a challenge in keeping inflation in check, the fiscal deficit on target and facilitating investment to ensure growth, while keeping expenditure on education and health on track. One hopes it will attain these targets.

The writer is chairman, Integrated Research and Action for Development (IRADE)

West Asia crisis has made India more energy secure



NIKITA AND BHARGABI GHOSH

ACCORDING TO data compiled by AQLin, 97 Indian cities featured among the world's top 100 hottest cities in May. Banda in Bundelkhand, Balangir in Odisha, Sasaram in Bihar and Varanasi in UP all hovered around 47-48°C. A few weeks before, the Union Cabinet approved India's revised Nationally Determined Contributions under the Paris Agreement, pledging to expand its carbon sink through forest and tree cover, increase non-fossil power capacity to 60 per cent, and reduce GDP emissions intensity by 47 per cent from 2005 levels by 2035. India has a strong record of meeting climate commitments. Yet the institutions dealing most di-

rectly with water stress, changing rainfall patterns and livelihood disruptions — gram panchayats — remain at the margins of climate planning.

Across rural India, climate change is reshaping everyday life. Farmers who once planned their sowing around fairly predictable monsoons now find themselves guessing when the rains will arrive and whether they will last long enough to sustain a crop. In many places, wells run dry before summer has ended and households are changing what they grow and eat. This is particularly consequential when set against rural-to-urban migration that is reshaping India's demographic geography.

It has been over three decades since the 73rd Amendment identified 29 subjects for devolution to gram panchayats, including water management, agriculture, social for-

estry and natural resource management, which sit at the centre of India's adaptation challenge today. The 16th Finance Commission, for the first time, integrated climate factors into its devolution recommendations. But the Panchayat Advancement Index 2.0 (PAI) and the Panchayat Devolution Index both show that functional devolution has declined in the same period. Without effective financial transfers, meaningful procedural authority, and genuine planning power, gram panchayats struggle to effectively respond to climate change.

There are many proven local models

When local representatives have opportunities to exchange what works, solutions travel faster

that demonstrate what becomes possible when gram panchayats are supported: Hiware Bazar in Maharashtra transformed a drought-prone landscape through watershed management, helped in part by lessons from neighbouring Ralegan Siddhi. A water conservation model from Bela in Maharashtra has direct relevance for a gram panchayat facing drought in Garhwal. The way a gram panchayat in Bokaro has used District Mineral Foundation funds for ecological restoration offers a working template for similar efforts in Chhattisgarh. Weather-risk management practices from Kerala's coastal gram panchayats can inform flood-prone communities in Assam. The Conference of Panchayats initiative, which has engaged more than 1,100 gram panchayats across five states, has shown that when local represen-

tatives have opportunities to exchange what works, solutions travel faster.

Three things would give ambitious climate targets a local foundation. First, the NAPCC and State Action Plans should formally bring gram panchayats in as planning partners with defined responsibilities and financial transfers. Second, PAI's nine thematic areas should connect high-performing panchayats with lagging ones in the same region, so that solutions move laterally rather than top down. Third, the 16th Finance Commission's climate-linked devolution must be paired with the procedural authority for gram panchayats to plan, not only to spend.

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• TECH

AI is reshaping warfare: How India can keep pace



EXPERT EXPLAINS
Lt GEN RAJ SHUKLA (RETD)
FORMER ARMY COMMANDER, UPSC MEMBER

THE TRINITY of artificial intelligence (AI), military autonomy and algorithmic warfare is redefining modern-day warfighting and deterrence. The workflows of combat are being infused with AI and driven by autonomy, but what is making them unfailingly lethal is algorithmic precision. Indeed, these three factors are poised to exponentially increase battlefield lethality over the next few years.

The strategic-military establishment in India would do well to take note of this 'Manhattan Project' moment in combat — the deployment of software at unprecedented speed and scale.

Tech in conflict: Ukraine, Venezuela and Iran

Drones are the greatest techno-military revolution since gunpowder and nuclear weapons. Their integration with AI and associated technologies has given birth to the art and science of drone warfare as warfare, even statecraft.

Their impact has been particularly visible in the Russia-Ukraine war. Ukraine, for instance, uses an AI-enabled data analytics platform called Delta that fuses multiple data inputs (from radar imagery to social media feeds) into one intelligent stream. Delta then plugs into a versatile inventory of drones to form a digital 'kill web' that compresses engagement times (detection to neutralisation) to a couple of minutes.

A 35-km corridor along the Russia-Ukraine frontier is nicknamed the "death zone". In this area, artillery has been pushed back and outranged, tanks don't move, and infantry soldiers are picked up by surveillance drones and neu-



Anduril's YFQ-44A began flight testing for the Collaborative Combat Aircraft Program in October 2025. ANDURIL

• From Ukraine to US: AI on the battlefield

UKRAINE'S DELTA battlefield management system has given it an edge in its grinding war against Russia. The system uses artificial intelligence to help with target selection, faster multi-sensor data analysis about enemy activity, and decision-making, among others.

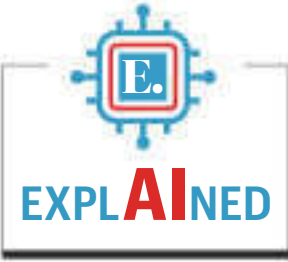
THE UKRAINIAN defence ministry

has said that Delta enables "real-time battlefield awareness, supports operational planning, and facilitates information sharing within units, brigades, formations, and, when needed, with allies".

ANOTHER EXAMPLE is YFQ-44A Fury, an AI-powered unmanned fighter jet developed by the American

defence tech startup Anduril.

ANDURIL SAYS on its website that integration of autonomy into fighter aircraft is a "step change" in aviation. "We designed YFQ-44A for a specific Air Force mission: to enhance survivability, lethality, and mission effectiveness by teaming with crewed fighter aircraft or operating independently."



A weekly column decoding AI's new frontiers, how they shape the world and our place in it

tralised by FPV (first-person view) drones in a matter of minutes. Unmanned platforms are taking over functions once performed by humans. This ranges from acquiring information and fixing targets to resupplying ammunition and rations, and evacuating casualties.

This year, Ukraine is buying eight million drones, more than the artillery it fired last year. These platforms provide commanders options ranging from close air support (25 km) to strategic air power (2,500 km).

In Venezuela, the US used Anthropic's Claude in the operation to capture ousted president Nicolas Maduro. AI platforms gave the US top brass insights into Maduro's movements and helped them anticipate what he would do.

Widening use cases

In Venezuela, the US used Anthropic's Claude in the operation to capture Maduro.

AI platforms gave the US top brass insights into Maduro's movements.

Electronic attacks and cyber exploits were then synchronised with the Delta Force's heliborne assault to telling effect.

In Iran, we saw the development of targeting packages at machine (not human) speeds, enabling precise strikes that wiped out almost the entire Iranian leadership in minutes on the morning of February 28.

Autonomous control

Military autonomy is making giant strides. Drone warfare is moving from remote to algorithmic control. The captains of naval platforms are increasingly going to be autonomous software systems, not humans. In the aerial domain, the development of the YFQ-44A Fury, an AI-powered autonomous fighter jet, by the defence tech startup Anduril, tells us that the future of airpower will be collaborative. This also shows that the business model is moving

Examples from Iran to Venezuela show that India needs to embrace the trinity of AI, autonomy and algorithmic warfare through sovereign pathways

steadily from traditional defence primes to agile startups.

Lethal technologies are seeing an unprecedented boom. Mythos is a virtual cyber-nuke that can disable any adversary operating system. DeepSeek and Tau's Law of Scaling (Huawei professing to achieve 1.4 nanometre transistor density by 2031 to challenge the Nvidia Blackwell chips which are in the range of 4 nanometres) are symbols of how Chinese digital innovations are turbo-charging strategic-military competition.

In combat theatres, software innovations are happening every three weeks, while new hardware is being deployed every three months. The Ministry of Defence, which has hitherto been a platform and weapons factory, will have to metamorphose into a software enterprise.

What India can do

The future of defence is also sovereignty. Beginning now, we can embrace the trinity of AI, autonomy and algorithmic warfare through these sovereign pathways:

- Urgently develop an AI-enabled data analytics platform, in the manner of Delta.
 - Create software that can autonomously coordinate drones swarms; identify objects of interest; differentiate between birds, civilian aircraft and combat platforms; and direct shooters to destroy targets.
 - Set up a diverse inventory of drones, with a target of five million by 2028.
 - Deploy an array of counter-drone systems (such as laser and microwave weapons) with drone-hunting teams. This will help to establish drone-driven, AI-enabled "kill webs" along the LoC and LAC.
 - Crowd the LEO (low-Earth orbit) space so that we transit from persistent surveillance to offensive ISR (intelligence, surveillance and reconnaissance).
 - The 2027 defence budget must reflect the transition. At least 40% of the Rs 2 lakh-odd crore earmarked for modernisation should be spent on technological solutions.
- This is a huge challenge that will require a cultural and structural transformation. Successful adoption, however, will be game changing. It will expand options in crisis situations while enabling India to attain "combat overmatch" with Pakistan and "asymmetric deterrence" with China.

• LAW

Inside West Bengal laws on property damage, preventive detention

Apurva Vishwanath
New Delhi, July 5

THE WEST Bengal Assembly recently passed two laws that mark the first major legislative move by the state's new BJP government: the West Bengal Public Safety and Control of Anti-social Activities Bill, 2026, and the West Bengal Maintenance of Public Order (Amendment) Bill, 2026.

Both cleared the House after a stormy debate, with the Public Safety Bill passing by 176 votes to 41. Chief Minister Suwendu Adhikari, the state's first BJP Chief Minister who also holds the Home portfolio, defended them as a solution to what he called the "goonda-necti" and "jungle raj" of 15 years of Trinamool rule.

The two laws follow a familiar template, which began in Uttar Pradesh and is now common in several BJP-ruled states: stringent laws and unusual state powers to recover damage to public property.

What is the new recovery law?

The West Bengal Maintenance of Public Order (Amendment) Bill, 2026 establishes a legal framework for claiming compensation for damage caused to public and private property during incidents like riots or anti-social activities. The proposed law seeks to amend the West Bengal Maintenance of Public Order Act, 1972 to create a money recovery mechanism for property destroyed during riots, arson, vandalism and protests.

How does the law operate?

The Bill follows the Uttar Pradesh format, where a 2020 order by district magistrates issued notices to individuals to recover the entire cost of damage to public property as assessed by the state government.

West Bengal's proposed law sets up a "Claims Commission" headed by a bureaucrat, an officer of at least Additional District Magistrate rank and armed with the powers of a civil court. After damage occurs, the District Magistrate or Police Commissioner must file a claim petition for losses to public property, while owners of damaged private property may file their own.

The Commission applies strict liability once a link between the incident and the damage is shown, can fix liability not only on those who caused the damage but on those who instigated, financed or harboured them, and may award "exemplary damages" of up to twice the compensation.

Compensation cannot fall below the property's market value, and unpaid amounts are recovered as arrears of land revenue, meaning property can be attached and auctioned. Crucially, every award is final, with no appeal to any court, and civil courts are barred from the subject matter.

After a Supreme Court rap, the UP government withdrew the Uttar Pradesh Recovery of Damages to Public and Private Property Act, 2020. In 2022, it enacted a new law, bringing in some independent

Stringent provisions

- After damage occurs, the District Magistrate or Police Commissioner must file a claim petition for losses to public property, while owners of damaged private property may file their own.
- The Commission applies strict liability once a link between the incident and the damage is shown, and can fix liability not only on those who caused the damage but on those who instigated, financed or harboured them.
- The proposed preventive detention law allows detention that cannot exceed 12 months, and also externment for up to a year.

former judges in the Claims Commission. Under the current law in UP, the tribunal is headed by a retired district judge. The new law is also under challenge before the SC.

The UP law grew directly out of the anti-CAA protests of December 2019.

The Yogi Adityanath government put up roadside hoardings naming and shaming those it accused of vandalism; the Allahabad High Court ordered the posters removed, and the Supreme Court observed that there was then no law to support the action. UP responded by promulgating an ordinance, later passed as an Act, setting up claims tribunals whose decisions could not be challenged in court.

In February 2022, the Supreme Court forced UP to withdraw 274 recovery notices and refund the money, holding that the State had acted as "complainant, adjudicator and prosecutor" in attaching the accused's properties, though it left the government free to start afresh through the statutory tribunal.

What is West Bengal's new preventive-detention law?

The Public Safety Bill is a standalone preventive detention statute aimed at "goondas" and "anti-social activities." Both terms are defined expansively: a goonda is defined as a person who is "generally reputed to be desperate and dangerous to the community" or is a "Habitual Offender."

An "anti-social activity" covers conduct that directly or indirectly causes "alarm or

insecurity among the public, danger to life or property, disturbance of public order, unlawful dispossession, or illegal mining, quarrying, sand or forest activity that drains the public exchequer."

West Bengal already has a preventive detention law dating back to 1970s. The present one is wider in its ambit.

Goonda laws exist in several states, most notably in Gujarat, as the Prevention of Anti-Social Activities Act, 1985. The Bharatiya Nyaya Sanhita also has provisions to deal with organised crime. Preventive detention is permitted under Article 22 of the Constitution with certain safeguards. Such issues are legislated under "public order," which is a state subject.

How does the preventive detention law operate?

The state government, or a district magistrate or police commissioner, once notified, may order a person detained to prevent such anti-social activity. Ordinarily, the person must have a conviction in the preceding seven years, or at least three separate charge-sheets.

Apart from a detention that cannot exceed 12 months, the proposed law allows externment, which entails removing a person from an area for up to a year.

It also allows for search and seizure, punishes the harbouring of a detained or externed person, grants officials good-faith immunity, and makes offences cognizable and non-bailable.

KOLKATA HAS THE GREATEST NUMBER OF UNINCORPORATED ENTERPRISES AT 8.84 LAKH

2 cr informal workers in India's big cities; Hyd tops with 15.7 lakh

Roughly 26% — or 52 lakh — of the informal workers in the country's most populous 46 cities are women

Siddharth Upasani
New Delhi, July 5

INFORMAL BUSINESSES are widely thought to operate largely in the rural heartland of the country. However, a new government report has shed light on the size of the informal sector in India's urban centres, estimating that the country's 46 most populous cities have 1.98 crore people working in 1 crore informal establishments.

Greater Hyderabad tops the list of cities with the highest number of informal workers at 15.7 lakh, with Kolkata home to the greatest number of informal establishments — 8.84 lakh.

As per the Ministry of Statistics and Programme Implementation's (MoSPI) report on urban unincorporated enterprises in cities whose population exceeded 1 million (10 lakh) as per Census 2011, six had more than a million informal workers: Greater Hyderabad, Delhi, Kolkata, Surat, Greater Mumbai, and Jaipur. Together, these six cities account for 40% of all informal workers in the country's 46 cities with more than a mil-

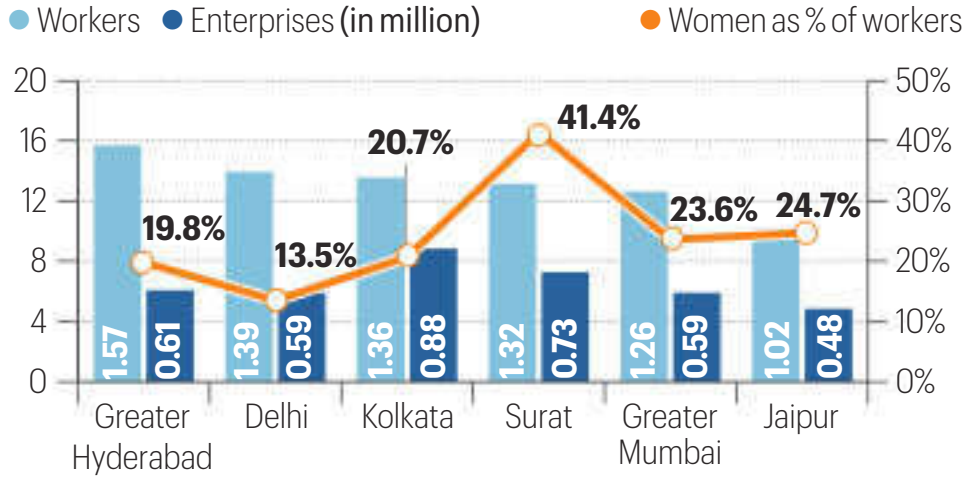
lion people each. The number of unincorporated establishments in these six cities is estimated to be nearly 39 lakh, or around 39% of the total in the 46 cities.

The total number of informal workers in India in 2025 was 12.81 crore.

Of the 1.98 crore informal workers in these 46 cities, roughly 26% — or 52 lakh — were women. Greater Visakhapatnam had the highest percentage of female workers at 42.5%, with Surat close behind at 41.4%. Srinagar was last at 10.5%, just below Varanasi at 12.1%. The report on the unincorporated enterprises in million-plus cities is the second such report released by MoSPI on India's biggest cities. The first was on their labour market dynamics, based on the results of the Periodic Labour Force Survey for 2025.

"This report presents, for the first time, a comprehensive statistical profile of 46 million-plus cities," Geeta Singh Rathore, Director General of MoSPI's National Sample Survey, said. She added that since the estimates are based on sample survey data, "differences observed across

• Cities with 1 million+ informal workers



NOTE: ESTIMATES ARE FOR UNINCORPORATED ENTERPRISES. SOURCE: MoSPI

cities may arise from variations in economic structure, industrial composition, local conditions and sampling variability".

The city-level estimates are generated from MoSPI's Annual Survey of Unincorporated Sector Enterprises (ASUSE) for 2025, which covers India's unincorporated, non-agricultural enterprises, excluding construction. These enterprises make up a sizable portion of the economy and include the likes of small manufacturers, service providers, and trading units.

Spread of cities

The 46 million-plus cities account for about 13% of establishments, 16% of workers, and 21% of Gross Value Added (GVA) of the unincorporated non-agricultural sector as per the ASUSE

2025 survey. These 46 cities are from 17 states and Union Territories: 10 from Maharashtra, seven from Uttar Pradesh, four each from Gujarat and Madhya Pradesh, three each from Tamil Nadu and Rajasthan, two each from Andhra Pradesh, Jharkhand, Punjab, and West Bengal, and one apiece from Bihar, Chhattisgarh, Delhi, Haryana, Jammu & Kashmir, Karnataka, and Telangana.

As *The Indian Express* had reported earlier this year in March, the number of informal establishments in the entire country rose by 58.5 lakh in 2025 to 792 crore, down from an increase of 83.5 lakh in 2023-24 (October-September). As a result, fewer jobs were created: 74.5 lakh in 2025 compared to 1.1 crore in the 12 months ended September

2024. Informal sector pay rose by just 3.9% in 2025, less than half the 13% increase seen in 2023-24.

Pay and productivity

In terms of total pay, emolument per worker was the highest in Jaipur, at Rs 2.33 lakh — although this only considers those unincorporated establishments with hired workers. Greater Hyderabad was second, at Rs 2.14 lakh. Emoluments refers to the total compensation a worker receives, inclusive of salary and benefits.

Meanwhile, informal enterprises in Gwalior paid hired workers the least at Rs 88,853, with Varanasi second from bottom at Rs 1 lakh. On average, emolument per worker in the 46 cities was Rs 1.5 lakh per year.

With the government releasing details for the 46 million-plus cities for the first time, comparative data on wages and the number of informal jobs is unavailable. Consequently, there is no way to calculate how much wages increased in 2025.

While informal workers in Jaipur got paid the most, they were not the most productive; that title belonged to Pimpri-Chinchwad, where the GVA per worker stood at Rs 2.9 lakh — significantly higher than Jaipur, which ranked 13th at Rs 2.23 lakh.

FULL REPORT ON
WWW.INDIANEXPRESS.COM

'Ethanol crucial for India; fastest, cheapest way to decarbonise road mobility'



VIKRAM GULATI
COUNTRY HEAD AND EVP,
TOYOTA KIRLOSKAR MOTORS

AS INDIA is witnessing transition in mobility both on account of different energy powertrains and ethanol blend of petrol, **Vikram Gulati**, country head and EVP, Toyota Kirloskar Motors told **Sandeep Singh** that all cleaner technologies need to be supported to help reduce carbon emissions, bring down pollution, and reduce dependence on imported energy. Amid growing concerns over impact of ethanol blend on fuel efficiency, he said the studies showed that the shift from E10 to E20 had negligible material impact and the impact on fuel efficiency was in the range of around 3-5%. He, however, said that for E25, new and older vehicles need to be evaluated before making that transition. He also called for greater policy support for hybrids for faster displacement of petrol and diesel cars. Edited excerpts:

Where do you see EVs, hybrids, ethanol-based flex-fuels fitting in the country's mobility transition?

If you look at India's priorities from a macro perspective, the objectives are very clear — reduce fossil fuel consumption, reduce carbon emissions, bring down pollution, increase domestic value addition and reduce dependence on imported energy.

I think at this stage, it makes sense to support all cleaner technologies. It is not EV versus hybrid — it is EVs and hybrids. Along with that, we need flex fuels, CNG, compressed biogas, hydrogen and other solutions working together.

The task is so huge that even if all these technologies grow, there will still be a gap and fossil fuels will continue to remain part of the system.

Anyone who wants to buy an EV will buy an EV. A hybrid is not going to distract that customer. But someone who is considering buying a petrol or diesel vehicle may think twice if a hybrid gives them the right total cost of ownership. So every hybrid sold is actually replacing a petrol or diesel vehicle. That helps the country, the city and the environment. Policy should ensure that the customer is not inadvertently taxed more for choosing such a clean technology.

Ethanol, on the other hand, is extremely relevant for India as

it is a domestic energy source, it supports the agricultural economy, helps reduce the amount of fossil fuel we consume and is the fastest as well as cheapest way to decarbonise road mobility. India's energy challenge is huge. There have been forecasts indicating that a significant part of the increase in global fossil fuel demand over the coming years will come from India because of economic growth and the increasing number of vehicles. The only way to address this is by improving efficiency and encouraging energy substitution through both electricity and ethanol.

India is pushing ethanol blending aggressively. Do you think this is the right time to accelerate the ethanol roadmap?

Definitely. If you look at the ethanol industry, there is significant available capacity today. The industry has been high-lighting that plants are operating below their potential, and there is enough opportunity to scale up ethanol production. This can help India reduce petrol consumption and fossil fuel dependence. This is the right time to move in that direction.

Brazil's experience is very important. Initially, after the oil crisis, they pushed ethanol-only vehicles. But customer adoption had limitations because people wanted flexibility. The major change happened when flex-fuel vehicles (FFVs) were introduced. FFVs were promoted by the government through enabling policies that encouraged rapid customer adoption. Once customers had the ability to use different ethanol blends depending on availability and economics, ethanol adoption increased sharply.

That is the real learning —

make the FFV technology customer-friendly, affordable to buy and run.

India has a strong opportunity. If policies create economic value for customers — both while buying flex-fuel vehicles and while using ethanol fuel — adoption can grow very quickly. For manufacturers also, scale matters. If volumes increase, engine supply chains can be developed for flex-fuel technology at a larger scale, which will help bring costs down.

The important point is that the transition has to be designed properly and the customer has to accept it. The technology, policy and economics all have to work together and there has to be value for the customer.

There are concerns around higher ethanol blends and their impact on fuel efficiency and engines. How do you see that?

We need to clearly differentiate between different blends. E85 is meant only for flex-fuel vehicles that are specifically designed for flexibly using any higher ethanol content. There was also some discussion on moving existing vehicles beyond E20, for example towards E25. That requires careful evaluation. The issue is not that it cannot be done but requires adequate testing to be completed and evaluated as was done earlier for E20.

When India moved from E10 to E20, a lot of studies were conducted. Older vehicles from different manufacturers were tested. Material compatibility, component deterioration and fuel efficiency impact were evaluated. The conclusion was that there was very negligible material impact. Fuel efficiency impact was minimal and estimated in the range of around 3-5%. Therefore, the concerns being seen in social media recently are not backed by scientific facts and sound technical evaluation. But for E25, similar extensive testing has not yet happened.

New vehicles and older vehicles need to be evaluated before making that transition. Testing will provide the answer. However, we feel that FFV provides agility and is the optimal way ahead.

FULL INTERVIEW ON
WWW.INDIANEXPRESS.COM

OPEC+ okays further oil output increase

Reuters
London, July 5

OPEC+ HAS agreed a further increase in output targets from August, the group said in a statement on Sunday, adding to global supply at a time when oil prices are falling due to the gradual reopening of the Strait of Hormuz for oil exports.

The oil-producing group agreed during an online meeting to increase quotas by 188,000 barrels per day (bpd) from August, on top of similar increases for June and July.

The seven core members of OPEC+, which groups OPEC and allied producers including Russia, have hiked their output quotas from April through July by almost 800,000 bpd. Yet the

increase has remained largely on paper because of the US-Israeli war on Iran, which closed the Strait of Hormuz to tanker traffic for some of the most important OPEC+ members, including Saudi Arabia, Kuwait and Iraq.

OPEC+ output fell to 33.13 million bpd in May, according to OPEC data, from 42.77 million bpd in February. It began to recover in June thanks to US efforts to help the UAE and other OPEC+ nations export more oil, but is still below pre-war levels.

Despite persisting supply disruptions, oil prices have returned to pre-war levels, pressured by lower Chinese imports, higher exports from non-Middle East producers, and a record global strategic stock re-



OPEC+ output fell to 33.13 million bpd in May, from 42.77 million bpd in February. REUTERS

lease coordinated by the International Energy Agency.

"The group of seven kept unwinding their production cuts as widely expected," UBS analyst

Giovanni Staunovo said. "The near-term focus will remain on how many tankers will manage to cross the Strait of Hormuz and how quickly demand and Chinese crude imports recover."

A memorandum of understanding between Washington and Tehran to end the war has also helped convince traders that supply will ultimately return to normal levels. Brent crude prices traded near \$72 per barrel on Friday, down from recent peaks of more than \$120 per barrel and back to levels traded just before the US and Israel attacked Iran on February 28.

Besides agreeing production targets, OPEC+ is also facing other challenges after the UAE left the group and Iraq signalled it wants higher quotas.